

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Software engineer with full-stack production experience across TypeScript, Python, and Java -- owning the complete SDLC from design through deployment at Isomer AI across 394 commits and 7 production deployments. Built REST APIs with role-based access control, CI/CD pipelines with GitHub Actions, and backend services with PostgreSQL, Redis, and Docker. Comfortable diagnosing and resolving production incidents, collaborating cross-functionally, and shipping software in iterative delivery cycles.

CORE COMPETENCIES

TypeScript + Node.js

Python + FastAPI

Java 21 + Spring Boot

REST APIs + RBAC

PostgreSQL + Redis

Docker + CI/CD

GitHub Actions

Agile SDLC

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Owned end-to-end SDLC for 10+ backend features on a production platform -- **394 commits and 7 production deployments**; diagnosed and resolved production incidents, participated in PR reviews, and maintained CI/CD pipelines with GitHub Actions
- Built REST APIs with Auth0 role-based access control, Temporal.io workflow orchestration, and real-time event pipelines using TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, and Docker
- Implemented document processing pipelines (PDF generation, Microsoft Graph API, S3 file storage) and multi-model AI tooling across Claude, GPT-4o, and Gemini with MCP and Sentry observability

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Designed and deployed a cloud-based genomic data platform on GCP Cloud Run enabling 100+ researchers to explore 500GB datasets; improved ML model accuracy by 12% through hyperparameter tuning

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool (OCR + NLP + ML); improved accuracy by 25%; automated AWS and GCP deployments with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led full SDLC for a support ticket analysis system, reducing processing time by 15%; distributed pipelines (Kafka + Spark) cut latency by 20%

PROJECTS

Inventory Reservation Service Java 21 + Spring Boot github.com/Aditya-k24/InventoryReservationService

Transactional inventory reservation API in Java 21 + Spring Boot with PostgreSQL row-level locking (SELECT FOR UPDATE) preventing overselling across multi-SKU reservations; idempotent POST endpoints with RFC 9457 error responses; automated expiry via scheduled cleanup job. Concurrency correctness verified with Testcontainers integration tests; Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI/CD.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Security, Testcontainers, Docker, GitHub Actions

PulseRank -- Real-Time Leaderboard Platform TypeScript + NestJS github.com/Aditya-k24/PulseRank

Event-driven leaderboard and activity feed: MySQL transactional writes with idempotency-key deduplication and outbox pattern, MongoDB feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads, Server-Sent Events for live updates. Multi-store consistency verified with Testcontainers; OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, GitHub Actions

AlgoPulse -- Agentic AI Platform TypeScript + Temporal github.com/Aditya-k24/AlgoPulse

Event-driven AI orchestration platform; diagnosed API-polling bottleneck under load and refactored to Supabase Edge Functions streaming AI reasoning loops to the client. Temporal.io and Kafka for state synchronization and fault tolerance at 2,000 async inference requests/second.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

VectorLift -- Semantic Search Engine Python + FastAPI + Kafka github.com/Aditya-k24/VectorLift

Production-grade semantic search system: Kafka-driven indexing pipeline, FastAPI serving with per-stage latency tracking, Prometheus + Grafana observability; 6 pipeline configurations evaluated with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure, OS

SKILLS

Languages: TypeScript, Python, Java, JavaScript, SQL

Backend: Node.js, Express, Spring Boot, FastAPI, NestJS, REST APIs, PostgreSQL, MySQL, MongoDB, Redis

DevOps: Docker, GitHub Actions CI/CD, GCP Cloud Run, AWS, Kubernetes, Terraform **Observability:** Prometheus, Grafana, Sentry, OpenTelemetry

AI/ML: Claude/OpenAI/Gemini APIs, MCP, LangChain, PyTorch, RAG, Temporal.io, Kafka